

HERMANN PAUWELLS RIVERA

Software Engineer | AI/ML · Full-Stack

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PROFILE

Software engineer focused on AI/ML: Python (TensorFlow, FastAPI) and full-stack TypeScript (React/Next.js, Node.js); final semester of a B.S. in Computer Science and Technology at Tecnológico de Monterrey, available full-time from January 2027. Founder of three early-stage startups built around LLMs and TypeSafe's Jev: AI-agent coordination, nutrition-label analysis and automated video production. Neural-network models for business cases (Datathon 2025).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Kotlin, C++
AI & ML: TypeSafe's Jev, LLMs (OpenRouter), TensorFlow, faster-whisper, Q-learning, AgentPy
Backend & Data: FastAPI, Flask, Node.js, PostgreSQL, Supabase, SQLite
Frontend & Tools: React, Next.js, Vite, Git, GitHub Actions, FFmpeg, Unity

EXPERIENCE

Full-Stack Developer (freelance), Juguetería El Arbolito Jun 2026 to present

- Built and shipped to production the online store and admin panel for a single-branch toy store with Next.js, TypeScript and Supabase (catalog, cart, checkout, backend).
- Integrated Mercado Pago payments and an LLM chatbot (OpenRouter); prepared the load of the 2,395-product catalog exported from the point of sale (Elevanta) and the online/in-store inventory sync (in progress).

STARTUPS (FOUNDER)

HIVEMIND Co-founder · team chat for people and AI agents · private pilot Sep 2026 to present

- Android (Kotlin/Compose) and Windows (Electron) apps over self-hosted ntfy; HMAC-SHA256 signed messages and 398 tests (148 Windows, 250 Android). Android 1.16: anti-replay and token scoped to the host.
- Bounded history and linear pending detection: from 2.5 s to 20 ms per event at 20,000 messages. Rust core in progress with Claude Code and Codex as agents; orchestration and context compaction with Jev.

AML consultancy Co-founder (technology) · anti-money-laundering · early stage Oct 2026 to present

- Co-founder in charge of technology at a Mexican AML compliance consultancy, with two professionals (one who designed a large retailer's AML system, one certified AML lawyer). Building the virtual office, brand, campaign and automated processes.
- Built the website with a vulnerable-activities self-assessment (LFPIORPI) and risk matrix, a brochure and the northwest campaign plan. Next: an AML system (matrices and manuals), as a license or service.

recaps-studio Co-founder · AI narrated-video factory (novel → manhwa-style video) Sep 2026 to present

- Resumable 13-stage Python pipeline with an orchestrator and live status page (TTS, AI images, FFmpeg/Remotion). Already published full-length videos on YouTube (the second runs 1 h 42 min); NVENC GPU encoding (~25% faster), MMS CTC forced alignment, a Spanish variant and distributed rendering.

Chécalo Founder · nutrition app for Mexico · in development Sep 2026 to present

- Scans barcodes and computes Mexico's NOM-051 warning labels with deterministic rules checked against the official gazette (DOF); TypeSafe's Jev gives an explainable verdict. iOS-style offline PWA and Android app (Kotlin); 53 tests.

SELECTED PROJECTS

CrediFiel Challenge (Datathon 2025) Data / ML · team project 2025

- Neural networks (regression + classification) to optimize Credifiel's direct-debit collections: minimize commissions or maximize recoveries.
- Real direct-debit collections dataset and KPIs; Flask API and a React + Vite frontend.

Multi-agent warehouse simulation (RL) Python · AgentPy · Unity · FastAPI 2024

- Multi-agent Q-learning with a blackboard (cell reservations, swap blocking) and Contract-Net task auctions: +105% deliveries per episode vs greedy allocation (20 seeded episodes).
- Animated Unity visualization and a FastAPI server.

Other projects: Lumina Reservations (WorkHub MTY) · folk-park · HeatShield · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

EDUCATION

B.S. in Computer Science and Technology (ITC), Tecnológico de Monterrey, Monterrey Campus 2021 to 2026
Expected graduation: December 2026.

AWARDS & ACTIVITIES

- Winner, Sustainability category, at Swift Challenge Fest 2025 (Tecnológico de Monterrey), with the HeatShield iOS app.
- SwiftTec Coding Community (2025): hackathons and innovation challenges.
- Music production and sound design (FL Studio).

Spanish (native) · English (C1) · German A2 (2-month intensive course in Heidelberg, Germany)